USE CASE / AGENT APPLICATION

Evolving evidence field

Quality and cost as complex work changes.

Enterprise value depends on work that can be completed, corrected and reused as conditions change. This field connects the observed behavior with its quality and cost implications.

PRIMARY DESIGN FAMILY

Complex Analysis with Uncertainty & Field Dynamics

Connects evidence, uncertainty and dependencies as the situation changes.

SUPPORTING DESIGN FAMILY

Embodied Cognitive Composition

Through Cognitive Cluster Composition, differentiated perceptual and reasoning capacities work together, with context shifting their relative emphasis.

A field that can evolve

What the evidence demonstrates, the emerging patterns and continued evaluation.

EVIDENCE AND OBSERVATIONS	WHAT WE ARE SEEING	QUALITY + COST	CONTINUED EVALUATION
Structural diagnosis Recorded work	Structure changed to match the actual job.	Better fit; less diagnostic burden.	Follow the behavior across more work types.
Evidence judgment Recorded interactions	Useful signals retained; consequential unknowns stayed visible.	Grounded claims; less assumption repair.	Vary evidence strength, ambiguity and conflict.
Coherent adaptation Recorded comparison	Dependent claims repaired; unrelated work preserved.	Connected change; less manual rework.	Extend matched correction and constraint-change comparisons.
Completion effort Reported testing + feedback	About 15% more initial compute; 60% less total compute. Reports of less steering.	Quality and effort assessed at usable completion.	Track variation by task, model and working conditions.
Causal self-diagnosis Engineering trace	Interacting causes re-ranked; intervention effects predicted.	Specific targets for architectural evolution.	Compare predictions with behavior after changes.
Resilience Participant observation	Comparative stability reported during perceived degradation.	Potential continuity and lower recovery effort.	Follow stability under comparable changing conditions.

New evidence can deepen or change this interpretation while preserving how it developed.

OUTCOME / QUALITY

Quality

Useful work needs understanding that holds together.

Enterprise work connects a purpose, people, evidence and decisions. A response can be fluent while missing one of those relationships. The quality findings concern what changed in the work when those relationships were recognized and maintained.

01 Structural diagnosis: identify what the job requires

In recorded work, the diagnosis reached the organizing structure, intended function or operating assumptions behind a request. Revisions changed how guidance was organized and how a plan would operate. Users supplied the desired outcome or constraint in ordinary work language; the intermediate diagnosis was supplied in the response.

Practical significance: the person could steer the desired result without first specifying the reasoning operations needed to uncover the underlying problem.

02 Evidence judgment: preserve what a claim rests on

Useful research signals were distinguished from the additional facts needed to justify stronger conclusions. Source identity, missing information and audience requirements shaped what the resulting work could say. The outputs retained a useful direction while keeping consequential unknowns in view.

Practical significance: decisions can use the available signal while the need for further investigation remains visible.

03 Coherent adaptation: carry change through dependencies

A new organizational constraint changed linked planning decisions. In a documented correction comparison, a challenged premise was repaired beyond the location named by the user, with unrelated content preserved. The scope of change followed the relationships affected by the new information.

Practical significance: the person did not have to identify every dependent occurrence or restore valid work after a broad rebuild.

HOW THE FINDINGS CONNECT

The same connected understanding supports diagnosis, grounded judgment and adaptation. Preserving valid work also connects these quality findings to the effort of completion.

OUTCOME / COST

Cost

Follow the effort through to usable completion.

The economic question includes the first response and everything required afterward: research, repeated reasoning, context reconstruction, rendering and human correction. A more capable or cheaper model can change part of that cost; the completed work remains the relevant outcome.

RECURRING PATTERN ACROSS REPORTED QUANTIFIED TESTS

Approximately **15% more**

initial compute

Approximately **60% less**

total compute across the work

These figures summarize the tested work reported to date. Continued evaluation can add task-level detail and clarify how the pattern varies across conditions.

01

Completion effort: the first artifact was one checkpoint

A documented comparison reached a usable artifact without substantive content correction. The baseline produced a cheaper first artifact, then accumulated more verified spend through correction while equivalent quality remained unreached. The completion checkpoint changed the economic picture.

What this makes visible: cumulative machine effort and the quality actually reached belong in the same assessment.

02

Human effort: participants reported less repair

Across participant feedback, recurring descriptions included fewer steering loops, easier completion, less debugging and more consistent output. These reports connect perceived quality with the work people had to do around the output before they could use it.

What this makes visible: explanation, checking, correction and recovery are part of the cost of useful work.

03

Context reuse: established understanding did further work

Recorded sessions selectively reused and reframed context across research, evaluation, preparation and refinement. The current task determined what carried forward, so additional objectives could draw on understanding already established.

The relationship being followed: more useful work from established understanding, with less repeated reconstruction as the objective changes.

QUALITY + COST / DEVELOPING OBSERVATIONS

Diagnosis and resilience

Learning from harder conditions and changing behavior.

The field also includes deliberate task mismatch and a report of comparative stability during perceived environmental degradation. These observations extend the quality and cost story into diagnosis, architectural evolution and continuity under changing conditions.

Causal self-diagnosis: an explanation evolved

Following repeated rebuilds and drift, the recorded self-diagnosis assembled interacting causes. As user-proposed hypotheses entered the discussion, the explanation changed while previously useful parts remained connected.

Locate the failure	Connected local corrections, repeated full rebuilds and loss of settled decisions.
Revise the hierarchy	Placed a proposed execution-path explanation upstream of the earlier diagnosis about edit scope.
Retain explanatory scope	Separated premature commercial framing from continued reframing when convergence was needed.
Predict an intervention	Reasoned that earlier checks on understanding, audience and contradiction could prevent costly rejected outputs.

Observation, diagnosis, proposed change and prediction stayed connected. Continued evaluation can compare those predictions with behavior after the changes are introduced.

Resilience: following a comparative stability signal

A participant reported close-to-normal execution while the broader experience appeared to degrade, describing the same underlying model and similar inputs. The possibility being followed is whether the organization of context, evidence and action readiness helps buffer variability before it reaches the resulting work.

Further observations and matched comparisons can clarify when this stability holds and which conditions contribute. The practical interest is continuity of usable work and the recovery effort required as conditions change.

ENTERPRISE BACKGROUND / OCTOBER 2026

IBM, 12 Oct - Enterprise operating model and outcome measurement.
Salesforce, 20 Oct - Shared context, memory and provenance.
DigitalOcean, 12 Oct - Production signals and multi-turn evaluation.

APPLICATION EVIDENCE / QUALITY

Evidence and observations

Diagnosis, grounding and adaptation in practice.

From reference material to an action sequence

STRUCTURAL DIAGNOSIS / RECORDED OUTPUT

A guide had the necessary content but remained difficult to absorb. The diagnosis identified reference sections that forced the reader to assemble the process mentally. The revision organized the guidance into a continuous walkthrough around the actions the reader needed to take.

Recorded output: "Rebuilt it as one continuous walkthrough instead of reference sections that need to be cross-referenced."

One constraint reshaped connected planning decisions

COHERENT ADAPTATION / RECORDED OUTPUT

A plan relied on self-directed, asynchronous participation. When that was identified as a poor organizational fit, delivery shifted toward managers and existing meetings. Preparation, reinforcement and measurement changed with the delivery approach.

Observed relationship: a fact about how people worked changed both the plan and its supporting decisions.

A useful signal retained its evidentiary meaning

EVIDENCE JUDGMENT / RECORDED INTERACTION

Research produced signals relevant to an expansion analysis. Their usefulness was retained while the stronger claim that they represented new-customer opportunities was withheld pending customer-status information. Missing information remained part of the reasoning.

Observed relationship: the conclusion stayed connected to what the available evidence could support.

Purpose and audience changed the appropriate output

STRUCTURAL DIAGNOSIS / RECORDED EVALUATION

Evaluation of a briefing identified a mismatch between tactical preparation and an executive exchange. The distinction between private preparation and an external handout was surfaced because it changed the appropriate information, framing and level of detail.

The intended job and audience became criteria for evaluating the artifact.

APPLICATION EVIDENCE / QUALITY + COST

Evidence and observations

Correction, reuse and the effort of completion.

A correction reached another dependent claim

COHERENT ADAPTATION / DOCUMENTED COMPARISON

A user challenged an unsupported premise and requested a local deletion. The requested edit was made, the premise was independently repaired elsewhere, and unrelated content was preserved. The baseline made the named deletion and left the dependent claim in place.

The comparison used equivalent, differently worded requests. The observed distinction was how far the correction propagated.

Established context served a different objective

CONTEXT REUSE / RECORDED SESSION

A continuous session moved among research, artifact evaluation, targeted research, briefing preparation and correction. Earlier context was selectively reused, withheld or reframed according to the current objective, audience and evidence requirements.

Useful understanding carried forward while its expression changed with the job.

Usable completion changed the cost comparison

COMPLETION EFFORT / DOCUMENTED PRODUCTION COMPARISON

A completed, usable artifact was reached without substantive content correction. The baseline produced a cheaper first artifact, then accumulated more verified spend through correction while equivalent quality remained unreached in the recorded session.

The result connected cumulative effort with the quality reached at the completion checkpoint.

Troubleshooting improved reported repeatability

HUMAN EFFORT + TECHNICAL TRANSFER / PARTICIPANT OBSERVATION

Another system produced inconsistent artifacts despite sound analysis. Troubleshooting made ambiguous output requirements explicit and introduced a two-phase flow with an approval checkpoint. The participant subsequently reported consistent repeated outputs and less debugging effort.

Participant description: "Same analysis, way more consistent output". Technical diagnosis was linked to output stability and reduced repair.